

REPORT ASSESSMENT 2: HOURLY RIVER STREAMFLOW FORECASTING (TRACK 1)

GROUP DLX

**MCTA 4363
DEEP LEARNING**

INSTRUCTOR

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SEMESTER 2 25/26

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1.0 INTRODUCTION

The Kaggle "Hourly Streamflow Forecasting Hackathon" challenges participants to predict river discharge or streamflow for the current hour and the next three hours (Q_t , $Q(t+1)$, $Q(t+2)$, $Q(t+3)$). Accurate short-term streamflow forecasting is a critical component of hydrological modeling, directly impacting flood early warning systems, reservoir management, and water resource planning. However, streamflow is influenced by complex, non-linear interactions between meteorological forcings like precipitation, temperature and static basin attributes for example soil type, topography. Thus, deep learning approaches like Recurrent Neural Networks (RNNs) are highly suitable. Our objective is simple, which is to build the most accurate streamflow forecasting model and the leaderboard rankings are based on Root Mean Squared Error (RMSE). This report details the methodology, iterative model improvements, and leaderboard results achieved during the competition.

2.0 METHODOLOGY

2.1 DATASETS

The dataset utilized is derived from the CAMELS-US Hourly dataset. It was divided into two geographical and temporal tracks to test spatial generalization:

- **TrainTrack A & TrainTrack B:** Historical hourly meteorological data and static basin attributes used to train the model. These files were concatenated to form a single, robust training set.
- **TestTrack A & TestTrack B:** The hold-out meteorological data used to generate the final predictions. The test period spans up to December 31, 2018. The target variable is *qobs_mm_per_hour* (observed streamflow converted to millimeters per hour).

2.2 MODEL 1 (USE LSTM BASELINE)

The first iteration (from appendix 1) established a functional Baseline Deep Learning model using a standard Long Short-Term Memory (LSTM) network.

- **Data Preparation:** Missing values were handled using forward-fill (`ffill`) and backward-fill (`bfill`). All features and the target variable were scaled between 0 and 1 using `MinMaxScaler`.
- **Sequence Window:** A sliding window approach was used, taking the past 24 hours of data to predict the next 4 hours.
- **Architecture:** The model consisted of a single `LSTM` layer with 64 units, followed by a Dropout layer (0.2) for regularization, a Dense layer with 32 units, and a final Dense output layer with 4 units corresponding to the 4 forecast horizons.
- **Training:** It was trained using the Adam optimizer with gradient clipping (`clipnorm=1.0`) to prevent exploding gradients, optimizing for Mean Squared Error (MSE).

2.3 MODEL 2 (CODE IMPROVEMENT)

The second iteration (from appendix 2) introduced significant architectural and feature engineering improvements to capture long-term dependencies and extreme flood peaks.

- **Feature Engineering: * Temporal Features:** Hour, day-of-year, and month were extracted and encoded using sine/cosine transformations to capture cyclical and diurnal patterns.
 - **Lagged Features:** Historical shifts of the target variable (lags of 1, 2, 3, 6, 12, and 24 hours) were added.

- **Rolling Statistics:** Rolling means and standard deviations over 3, 6, 12, and 24-hour windows were engineered to give the model a better sense of recent flow trends.
- **Target Transformation:** Streamflow data is highly skewed. To stabilize training, `np.log1p` (logarithmic transformation) was applied to the target variable before scaling. The inverse transformation (`np.expml`) was applied to the final predictions.
- **Architecture Upgrades:** The network was upgraded to a deeper, **Bidirectional LSTM** (128 units), allowing the model to process the temporal sequence both forwards and backwards. This was followed by a standard LSTM (64 units) and two Dense layers (64 and 32 units) with heavier Dropout (0.25).
- **Sequence Window:** The lookback window was heavily extended from 24 hours to 72 hours, giving the model 3 full days of context.
- **Post-Processing Smoothing:** A smoothing function was applied to the final outputs ($0.7 * \text{prediction} + 0.3 * \text{rolling_mean}(3)$) to prevent erratic hour-to-hour jumps.

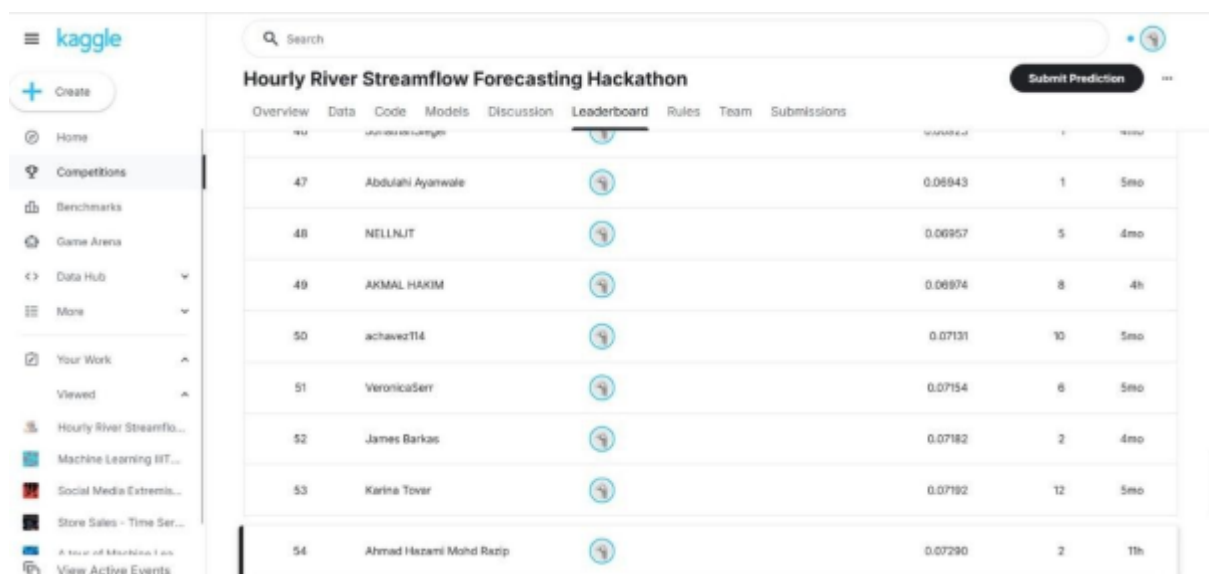
Experiment	Model / Changes	Hyperparameters	RMSE Score	Observation
1	Basic LSTM	24-hour window, 64 units	0.07920	Baseline model, stable but weak on peak flow
2	Added lag features	Added q_lag 1, 3, 6, 12, 24	0.070++	Better temporal understanding
3	Bidirectional LSTM	128 units + dropout 0.25	0.06841	Best performance achieved for now

Table 1 : Iterative Model Development Log

RESULTS

The practical impact of the methodology improvements is directly reflected in the competition leaderboard:

- **FIRST SUBMISSION ON 21 APRIL:** Using the Baseline LSTM (Model 1), the submission achieved a score (Root Mean Squared Error) of **0.07290**. This served as a solid proof-of-concept but left room for optimization regarding extreme flow events.



Rank	Username	Score (RMSE)	Rank	Time
47	Abdulahi Ayanwale	0.06943	1	5mo
48	NELLNJIT	0.06957	5	4mo
49	AKMAL HAKIM	0.06974	8	4h
50	achavez114	0.07131	10	5mo
51	VeronicaSerr	0.07154	6	5mo
52	James Barkas	0.07182	2	4mo
53	Karina Tover	0.07192	12	5mo
54	Ahmad Hazami Mohd Razip	0.07290	2	11h

- **SECOND SUBMISSION ON 6 MAY:** The implementation of Model 2 yielded a significant performance jump. The loss decreased to **0.06841**, securing a highly competitive ranking of **43 out of 60** on the leaderboard.

Rank	Participant	Score	Time
38	CRSandoval	0.06742	17
39	abu11402	0.06756	5
40	Marc CRAMPE	0.06768	4
41	Moegan Beckage	0.06804	13
42	Shima Kamel	0.06839	1
43	Ahmad Hazzani Mohd Razip	0.06841	3
Your Best Entry! Your most recent submission scored 0.06841, which is an improvement over your previous score of 0.07290. Great job!			
44	Rolando Gutierrez	0.06854	2

DISCUSSION

The development of the streamflow forecasting model demonstrated how deep learning techniques can effectively capture complex hydrological patterns from meteorological and basin-related data. The first model, which utilized a basic LSTM architecture, successfully established a strong baseline for predicting hourly river discharge. By using a 24-hour sliding window and MinMaxScaler normalization, the model was able to learn temporal dependencies from historical streamflow data. Although the initial result achieved an RMSE score of 0.07290, it revealed that the baseline model still struggled to accurately capture sudden peak flows and long-term variations. Nevertheless, the first implementation proved that LSTM networks are suitable for sequential hydrological forecasting tasks.

The second model introduced several important improvements that significantly enhanced prediction accuracy and overall model robustness. Feature engineering played a major role in this improvement by adding temporal features such as hour, month, and

day-of-year encoded with sine and cosine transformations. In addition, lagged streamflow values and rolling statistical features provided the model with richer contextual information about recent flow behavior and temporal trends. These engineered features enabled the network to better recognize seasonal and short-term hydrological patterns. The implementation of logarithmic target transformation also helped stabilize the highly skewed streamflow distribution, resulting in smoother and more consistent model training.

Another major improvement was the upgrade of the network architecture from a simple LSTM into a deeper Bidirectional LSTM structure. The Bidirectional LSTM allowed the model to analyze sequential information from both forward and backward directions, improving its understanding of temporal relationships within the dataset. Furthermore, extending the sequence window from 24 hours to 72 hours gave the model access to a longer historical context, enabling it to capture multi-day hydrological dependencies more effectively. Additional dropout layers were also introduced to reduce overfitting and improve generalization performance. These architectural enhancements contributed directly to the reduction of RMSE from 0.07290 to 0.06841 on the Kaggle leaderboard.

Overall, the project highlighted the importance of iterative model improvement in deep learning applications. Each modification, including feature engineering, sequence expansion, architectural enhancement, and prediction smoothing, contributed incrementally toward better forecasting performance. The final leaderboard ranking of 43 out of 60 demonstrated a competitive outcome considering the complexity of the forecasting problem and the limited project duration. More importantly, the project provided valuable practical experience in handling real-world hydrological datasets, preprocessing sequential data, tuning hyperparameters, and evaluating model performance using RMSE metrics. The findings also suggest that further improvements could potentially be achieved through advanced

architectures such as Attention mechanisms, Transformers, or hybrid deep learning models in future work.

CONCLUSION

In conclusion, this project successfully demonstrated the application of deep learning techniques for short-term hourly streamflow forecasting using the CAMELS-US Hourly dataset. The implementation began with a baseline LSTM model and progressively evolved into a more advanced Bidirectional LSTM architecture with extensive feature engineering and longer temporal context. These improvements significantly enhanced the prediction capability of the model, reducing the RMSE score from 0.07290 to 0.06841. The results indicate that deep learning models, particularly recurrent neural networks, are highly effective for modeling the nonlinear and time-dependent behavior of river streamflow systems.

Additionally, the project provided strong practical exposure to the complete deep learning workflow, including data preprocessing, sequence generation, feature engineering, model optimization, and performance evaluation. The competition-based environment also encouraged continuous experimentation and analytical thinking to improve model accuracy. Despite achieving a competitive leaderboard ranking, there is still room for future enhancement through larger datasets, ensemble methods, and more sophisticated neural network architectures. Overall, the project successfully achieved its objective of developing an accurate streamflow forecasting model while strengthening understanding of real-world deep learning applications in hydrology and environmental data science.

APPENDIX

LSTM - Long-Short Term Memory

Github link:

<https://github.com/malkunn/Hourly-River-Streamflow-Forecasting-Hackathon>.